Board Sheet — Lesson 9: Linear Regression II

MA153X Mathematical Modeling

Name: ______________________________ **Date:** ____________________

Record Your Model from Lesson 8

$$\hat{y} = \_\_\_\_\_\_\_\_\_ + \_\_\_\_\_\_\_\_\_ \cdot x$$

where $x =$ ____________________ and $\hat{y} =$ ____________________

Interpreting the Slope ($\hat{\beta}_1$)

Template (Reading 2.2): "For each one-inch increase in ______________, the model predicts a(n) [increase / decrease] of _______ ______________."

Write your interpretation of the slope in context:

Caution: Regression shows ______________, not causation!

Interpreting the Intercept ($\hat{\beta}_0$)

Template: "When ______________ is zero, the model predicts ______________ is _________ [units]."

Is this meaningful in context? Why or why not?

What is extrapolation? ______________________________

Interpreting R^2

Record R^2 from Excel: `=RSQ(...)` $\rightarrow$ ________________

Template: "Approximately ________% of the variability in __________________ is explained by the linear relationship with __________________."

R^2 does **NOT** tell you (check all that apply):

- ☐ Whether the model is appropriate for the data
- ☐ Whether the model predicts well on new data
- ☐ Whether the relationship is causal

Out-of-Sample Predictions

Using the test data, predict weight for each soldier:

Soldier	Height (in)	Actual Weight (lbs)	Predicted ($\hat{y}$)	Error ($y - \hat{y}$)	|Error|
S001	66	203			
S016	66	189			
S018	66	172			
S031	62	157			
S034	68.5	149			
S040	66	166			
S042	64	147			
S044	71	184			
S046	61	127			
S047	67	172			

Excel: Predicted = `= intercept_cell + slope_cell * Height_cell`

Mean Absolute Error (MAE)

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i|$$

Metric	Training	Test
MAE		
n	37	10

Model Validation Checklist (from Reading 2.2)

Check	Result	Pass?
Scatter plot shows linear trend?		[ ]
Training R^2 reasonable?	$R^2 =$	[ ]
Test MAE $\approx$ Training MAE?	Train: _____ Test: _____	[ ]
Predictions reasonable in context?		[ ]

Decision

Based on the evidence, should we **support** or **question** this model? Justify:

Takeaways

What does the slope tell you in this problem?

Why do we validate on test data instead of just using training R^2?
